

Education

Computer Science and Technology **Sun Yat-Sen University** **Aug 2023 - June 2027(Expected)**

- **Coursework:** Artificial Intelligence; Machine Learning; Optimization Theory; Data Structures and Algorithms; Mathematical analysis I & II;
- **GPA:** 3.8/4.0

Research & Project Experience

Estimate Relative Camera Pose **Sep 2025 – Feb 2026**

- Designed a minimal solver for relative pose estimation using point correspondences and surface normals.
- Evaluated robustness and efficiency with several benchmarks.

Optimization of code_saturne **ISC Student Cluster Competition** **March 2025 – May 2025**

- Addressed communication bottlenecks via MPI implementation comparisons and disk I/O minimization strategies.
- Achieved a significant 2.7 times performance improvement for the LES_AIR_QUALITY case on Bridges 2.
- Fixed GPU runtime errors; scalability requires architectural redesign.

vLLM-Ascend Sampler **Nov 2025 – Jan 2026**

- Identified performance bottlenecks in Top-K / Top-P sampling during LLM inference on Ascend NPUs.
- Implemented Triton-based fused sampling kernels to reduce kernel launches and memory access overhead.
- Achieved up to 1.88% latency reduction compared to the native Ascend 910 implementation.

Optimization of HPL-MxP **Nov 2025 – Feb 2026**

- Evaluating performance and reliability of Huawei SDMA for large-scale dense linear algebra.
- Integrating SDMA support into the HPL-MxP execution framework.
- Optimizing single-block low-precision LU factorization without panel reordering.

Yat-CC **Feb. 2025 – July 2025**

- Refined the Intelligent TA knowledge base by optimizing data structures for enhanced retrieval quality.
- Curated a compiler theory instruction-tuning dataset sourced from lecture slides and course materials.
- Applied PEFT (LoRA) to Qwen3-Instruct-8B via PyTorch, boosting code comprehension and error feedback logic.

Awards

- ISC 2025 Student Cluster Competition 1st Place Online Winner
- 2023-2025 Sun Yat-Sen University Scholarship

Languages and Technologies

- **Languages:** English (CET-6: 544); Mandarin (Native); Cantonese (Native)

- **Programming:** Python (familiar); C/C++ (familiar); OpenMP (familiar); MPI (working knowledge); CUDA (learning)
- **Systems & Tools:** Linux; GDB; CMake; Performance profiling and benchmarking; Docker